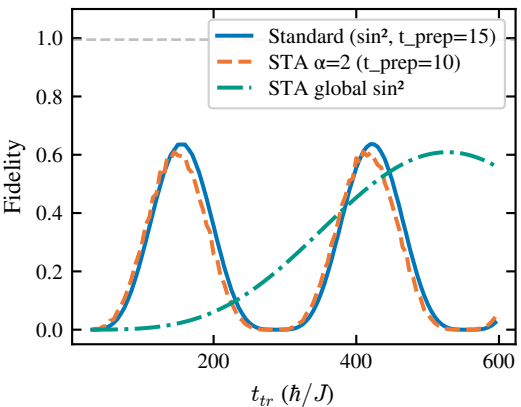


Combined study: asymmetric position + Shortcut to Adiabaticity

$$L = 21, v_{tr} = 0.5, w = 1.0$$

DW position	Site	Results summary: DW position $\times$ protocol	t_{tr} opt.	Fidelity
Center ($\sim 1/2$)	DW=11	Standard ($\sin^2$, $t_{\text{prep}}=15$)	420.0	0.6368
Center ($\sim 1/2$)	DW=11	STA $\alpha=2$ ($t_{\text{prep}}=10$)	410.0	0.6099
Center ($\sim 1/2$)	DW=11	STA global $\sin^2$	530.0	0.6089
Third ($\sim 1/3$)	DW=7	Standard ($\sin^2$, $t_{\text{prep}}=15$)	95.0	0.0606
Third ($\sim 1/3$)	DW=7	STA $\alpha=2$ ($t_{\text{prep}}=10$)	380.0	0.0597
Third ($\sim 1/3$)	DW=7	STA global $\sin^2$	270.0	0.0474
Quarter ($\sim 1/4$)	DW=5	Standard ($\sin^2$, $t_{\text{prep}}=15$)	495.0	0.0040
Quarter ($\sim 1/4$)	DW=5	STA $\alpha=2$ ($t_{\text{prep}}=10$)	200.0	0.0047
Quarter ($\sim 1/4$)	DW=5	STA global $\sin^2$	120.0	0.0025

(b) DW center (site 11): protocol comparison



(c) DW quarter (site 5): protocol comparison

